

How does this implementation (V2) differ from the traditional DSW approach (V1).

- What is the impact of phase 1: In V1, we first rotate **every right child** to make a left-skewed vine, which gives a similar starting point no matter the original shape of a tree. In V2, we get different starting points for different trees as we rotate only nodes with subtrees of height > 2 to reduce the number of rotations.
- What is the impact of phase 2: In V1, we target a $\log_2 n$ height from a vine with n nodes. In V2, we rotate until the height is double that of a balanced tree ($2\log_2 n$)
- What tradeoff did you observe: In phase 2, for example, the resulting balanced trees have larger heights but take less rotations to get to.
- Can you think of other variations. Explain this in detail and justify your design choices.
 - Skipping phase 1 but still following V2 i.e. rotating only when right children > 2 . It looks possible to balance at least a subtree as in the example below. This process saves resources used in phase 1 while accomplishing some level of balancing in phase 2.

